Table 15 WHO base anthropometric outcomes for primary school, youngest child

Current school enrollment defines the primary school sample. One youngest child per household is selected within this level. WHO 2007 sex-specific monthly growth references use age on 16 December 2022. WAZ is defined only through month 120. Father and mother reform estimates use the established weighted RDD.

Father RDD primary analysis

Specification 1 Monthly child-age fixed effects No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Height-for-age z score	-0.280	0.033 [0.198] N=1,971	0.095 [0.207] N=1,819	0.189 [0.215] N=1,638	0.108 [0.229] N=1,432	0.212 [0.250] N=1,199
Weight-for-age z score, through 120 months	0.300	0.001 [0.146] N=1,986	0.007 [0.157] N=1,834	0.079 [0.168] N=1,652	0.078 [0.181] N=1,440	0.114 [0.201] N=1,206
BMI-for-age z score	0.559	-0.010 [0.168] N=1,934	-0.021 [0.177] N=1,781	0.037 [0.186] N=1,605	0.046 [0.206] N=1,403	0.085 [0.223] N=1,176
Stunted, HAZ below -2	0.176 P=339	-0.032 [0.036] N=1,971	-0.036 [0.038] N=1,819	-0.050 [0.040] N=1,638	-0.043 [0.042] N=1,432	-0.070 [0.047] N=1,199
Underweight, WAZ below -2, through 120 months	0.048 P=96	**0.038* [0.021] N=1,986**	**0.046** [0.022] N=1,834**	0.036 [0.023] N=1,652	**0.057** [0.023] N=1,440**	**0.065*** [0.024] N=1,206**
Thinness, BMIZ below -2	0.085 P=165	0.036 [0.027] N=1,934	0.030 [0.028] N=1,781	0.027 [0.030] N=1,605	**0.050* [0.029] N=1,403**	0.038 [0.034] N=1,176
Overweight including obesity, BMIZ above +1	0.406 P=789	0.046 [0.049] N=1,934	0.030 [0.053] N=1,781	0.039 [0.056] N=1,605	0.067 [0.062] N=1,403	0.065 [0.066] N=1,176
Obesity, BMIZ above +2	0.219 P=420	0.052 [0.037] N=1,934	0.034 [0.038] N=1,781	0.062 [0.041] N=1,605	**0.073* [0.043] N=1,403**	0.046 [0.047] N=1,176

Specification 1 Monthly child-age fixed effects 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Height-for-age z score	-0.392 [0.300] N=1,658	-0.350 [0.329] N=1,506	-0.246 [0.361] N=1,325	-0.571 [0.398] N=1,119	-0.669 [0.503] N=886
Weight-for-age z score, through 120 months	-0.132 [0.190] N=1,668	-0.138 [0.216] N=1,516	-0.033 [0.241] N=1,334	-0.132 [0.275] N=1,122	-0.192 [0.341] N=888
BMI-for-age z score	0.082 [0.242] N=1,629	0.076 [0.269] N=1,476	0.197 [0.293] N=1,300	0.245 [0.347] N=1,098	0.331 [0.446] N=871
Stunted, HAZ below -2	0.060 [0.052] N=1,658	0.069 [0.056] N=1,506	0.058 [0.061] N=1,325	0.100 [0.065] N=1,119	0.100 [0.080] N=886
Underweight, WAZ below -2, through 120 months	0.035 [0.034] N=1,668	0.052 [0.036] N=1,516	0.028 [0.042] N=1,334	**0.081* [0.043] N=1,122**	**0.121** [0.048] N=888**
Thinness, BMIZ below -2	0.017 [0.037] N=1,629	0.002 [0.041] N=1,476	-0.008 [0.044] N=1,300	0.028 [0.047] N=1,098	0.010 [0.067] N=871
Overweight including obesity, BMIZ above +1	0.062 [0.071] N=1,629	0.036 [0.081] N=1,476	0.054 [0.091] N=1,300	0.112 [0.108] N=1,098	0.121 [0.134] N=871
Obesity, BMIZ above +2	0.080 [0.053] N=1,629	0.052 [0.058] N=1,476	**0.106* [0.063] N=1,300**	**0.144** [0.068] N=1,098**	0.085 [0.081] N=871

Specification 2 Monthly fixed effects and female No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Height-for-age z score	-0.280	0.034 [0.196] N=1,971	0.095 [0.206] N=1,819	0.190 [0.215] N=1,638	0.108 [0.228] N=1,432	0.205 [0.246] N=1,199
Weight-for-age z score, through 120 months	0.300	0.001 [0.144] N=1,986	0.008 [0.155] N=1,834	0.083 [0.166] N=1,652	0.076 [0.179] N=1,440	0.101 [0.197] N=1,206
BMI-for-age z score	0.559	-0.007 [0.170] N=1,934	-0.017 [0.178] N=1,781	0.044 [0.187] N=1,605	0.046 [0.207] N=1,403	0.078 [0.225] N=1,176
Stunted, HAZ below -2	0.176 P=339	-0.032 [0.036] N=1,971	-0.036 [0.038] N=1,819	-0.050 [0.040] N=1,638	-0.043 [0.042] N=1,432	-0.070 [0.046] N=1,199
Underweight, WAZ below -2, through 120 months	0.048 P=96	**0.038* [0.021] N=1,986**	**0.046** [0.022] N=1,834**	0.036 [0.023] N=1,652	**0.057** [0.023] N=1,440**	**0.065*** [0.024] N=1,206**
Thinness, BMIZ below -2	0.085 P=165	0.036 [0.027] N=1,934	0.031 [0.028] N=1,781	0.028 [0.030] N=1,605	**0.050* [0.030] N=1,403**	0.038 [0.034] N=1,176
Overweight including obesity, BMIZ above +1	0.406 P=789	0.047 [0.050] N=1,934	0.031 [0.053] N=1,781	0.041 [0.057] N=1,605	0.067 [0.062] N=1,403	0.063 [0.066] N=1,176
Obesity, BMIZ above +2	0.219 P=420	0.053 [0.037] N=1,934	0.035 [0.039] N=1,781	0.064 [0.041] N=1,605	**0.073* [0.044] N=1,403**	0.043 [0.048] N=1,176

Specification 2 Monthly fixed effects and female 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Height-for-age z score	-0.392 [0.301] N=1,658	-0.352 [0.330] N=1,506	-0.253 [0.363] N=1,325	-0.575 [0.400] N=1,119	-0.667 [0.503] N=886
Weight-for-age z score, through 120 months	-0.123 [0.190] N=1,668	-0.129 [0.217] N=1,516	-0.019 [0.242] N=1,334	-0.124 [0.275] N=1,122	-0.201 [0.337] N=888
BMI-for-age z score	0.094 [0.244] N=1,629	0.091 [0.271] N=1,476	0.227 [0.296] N=1,300	0.260 [0.348] N=1,098	0.333 [0.446] N=871
Stunted, HAZ below -2	0.061 [0.052] N=1,658	0.070 [0.056] N=1,506	0.059 [0.062] N=1,325	0.101 [0.065] N=1,119	0.100 [0.081] N=886
Underweight, WAZ below -2, through 120 months	0.035 [0.034] N=1,668	0.052 [0.037] N=1,516	0.028 [0.042] N=1,334	**0.080* [0.043] N=1,122**	**0.121** [0.048] N=888**
Thinness, BMIZ below -2	0.018 [0.037] N=1,629	0.002 [0.041] N=1,476	-0.007 [0.044] N=1,300	0.029 [0.047] N=1,098	0.010 [0.067] N=871
Overweight including obesity, BMIZ above +1	0.066 [0.071] N=1,629	0.041 [0.081] N=1,476	0.063 [0.091] N=1,300	0.117 [0.109] N=1,098	0.122 [0.134] N=871
Obesity, BMIZ above +2	0.084 [0.054] N=1,629	0.058 [0.058] N=1,476	**0.114* [0.064] N=1,300**	**0.149** [0.068] N=1,098**	0.085 [0.082] N=871

Specification 3 Monthly fixed effects female and household income groups No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Height-for-age z score	-0.280	0.016 [0.195] N=1,971	0.062 [0.205] N=1,819	0.146 [0.214] N=1,638	0.035 [0.226] N=1,432	0.117 [0.245] N=1,199
Weight-for-age z score, through 120 months	0.300	-0.011 [0.144] N=1,986	-0.011 [0.155] N=1,834	0.061 [0.166] N=1,652	0.050 [0.179] N=1,440	0.073 [0.197] N=1,206
BMI-for-age z score	0.559	-0.007 [0.170] N=1,934	-0.012 [0.178] N=1,781	0.057 [0.186] N=1,605	0.077 [0.206] N=1,403	0.130 [0.223] N=1,176
Stunted, HAZ below -2	0.176 P=339	-0.030 [0.037] N=1,971	-0.032 [0.039] N=1,819	-0.044 [0.041] N=1,638	-0.034 [0.043] N=1,432	-0.059 [0.047] N=1,199
Underweight, WAZ below -2, through 120 months	0.048 P=96	**0.039* [0.021] N=1,986**	**0.048** [0.022] N=1,834**	0.037 [0.024] N=1,652	**0.060** [0.024] N=1,440**	**0.068*** [0.025] N=1,206**
Thinness, BMIZ below -2	0.085 P=165	0.036 [0.027] N=1,934	0.029 [0.029] N=1,781	0.024 [0.030] N=1,605	0.043 [0.030] N=1,403	0.028 [0.033] N=1,176
Overweight including obesity, BMIZ above +1	0.406 P=789	0.047 [0.050] N=1,934	0.031 [0.053] N=1,781	0.042 [0.056] N=1,605	0.072 [0.061] N=1,403	0.073 [0.066] N=1,176
Obesity, BMIZ above +2	0.219 P=420	0.051 [0.038] N=1,934	0.033 [0.039] N=1,781	0.061 [0.041] N=1,605	**0.074* [0.044] N=1,403**	0.047 [0.049] N=1,176

Specification 3 Monthly fixed effects female and household income groups 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Height-for-age z score	-0.417 [0.292] N=1,658	-0.412 [0.319] N=1,506	-0.337 [0.351] N=1,325	**-0.743* [0.380] N=1,119**	**-0.923* [0.474] N=886**
Weight-for-age z score, through 120 months	-0.138 [0.189] N=1,668	-0.160 [0.216] N=1,516	-0.057 [0.241] N=1,334	-0.180 [0.274] N=1,122	-0.289 [0.336] N=888
BMI-for-age z score	0.092 [0.244] N=1,629	0.096 [0.271] N=1,476	0.244 [0.292] N=1,300	0.322 [0.344] N=1,098	0.446 [0.432] N=871
Stunted, HAZ below -2	0.066 [0.050] N=1,658	0.079 [0.053] N=1,506	0.074 [0.059] N=1,325	**0.128** [0.062] N=1,119**	**0.138* [0.076] N=886**
Underweight, WAZ below -2, through 120 months	0.036 [0.034] N=1,668	0.054 [0.037] N=1,516	0.031 [0.043] N=1,334	**0.086* [0.044] N=1,122**	**0.130*** [0.049] N=888**
Thinness, BMIZ below -2	0.017 [0.036] N=1,629	-0.000 [0.040] N=1,476	-0.014 [0.043] N=1,300	0.014 [0.047] N=1,098	-0.011 [0.066] N=871
Overweight including obesity, BMIZ above +1	0.065 [0.071] N=1,629	0.040 [0.081] N=1,476	0.063 [0.091] N=1,300	0.120 [0.108] N=1,098	0.134 [0.132] N=871
Obesity, BMIZ above +2	0.082 [0.054] N=1,629	0.056 [0.058] N=1,476	**0.110* [0.064] N=1,300**	**0.151** [0.068] N=1,098**	0.098 [0.083] N=871

Specification 4 Monthly fixed effects female and per-person income groups No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Height-for-age z score	-0.280	0.004 [0.194] N=1,971	0.039 [0.205] N=1,819	0.125 [0.213] N=1,638	0.025 [0.224] N=1,432	0.093 [0.245] N=1,199
Weight-for-age z score, through 120 months	0.300	-0.013 [0.141] N=1,986	-0.018 [0.153] N=1,834	0.050 [0.164] N=1,652	0.040 [0.177] N=1,440	0.058 [0.195] N=1,206
BMI-for-age z score	0.559	-0.001 [0.169] N=1,934	-0.005 [0.177] N=1,781	0.059 [0.186] N=1,605	0.071 [0.206] N=1,403	0.120 [0.221] N=1,176
Stunted, HAZ below -2	0.176 P=339	-0.026 [0.036] N=1,971	-0.025 [0.038] N=1,819	-0.038 [0.041] N=1,638	-0.029 [0.042] N=1,432	-0.050 [0.047] N=1,199
Underweight, WAZ below -2, through 120 months	0.048 P=96	**0.038* [0.021] N=1,986**	**0.048** [0.022] N=1,834**	0.038 [0.024] N=1,652	**0.061** [0.023] N=1,440**	**0.069*** [0.024] N=1,206**
Thinness, BMIZ below -2	0.085 P=165	0.034 [0.027] N=1,934	0.027 [0.029] N=1,781	0.024 [0.030] N=1,605	0.044 [0.030] N=1,403	0.027 [0.034] N=1,176
Overweight including obesity, BMIZ above +1	0.406 P=789	0.048 [0.050] N=1,934	0.033 [0.053] N=1,781	0.044 [0.057] N=1,605	0.074 [0.062] N=1,403	0.074 [0.066] N=1,176
Obesity, BMIZ above +2	0.219 P=420	0.053 [0.038] N=1,934	0.036 [0.039] N=1,781	0.065 [0.041] N=1,605	**0.078* [0.043] N=1,403**	0.050 [0.048] N=1,176

Specification 4 Monthly fixed effects female and per-person income groups 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Height-for-age z score	-0.407 [0.296] N=1,658	-0.422 [0.326] N=1,506	-0.349 [0.358] N=1,325	**-0.726* [0.387] N=1,119**	**-0.940* [0.501] N=886**
Weight-for-age z score, through 120 months	-0.130 [0.185] N=1,668	-0.160 [0.213] N=1,516	-0.062 [0.238] N=1,334	-0.184 [0.269] N=1,122	-0.312 [0.333] N=888
BMI-for-age z score	0.089 [0.242] N=1,629	0.099 [0.269] N=1,476	0.248 [0.291] N=1,300	0.309 [0.340] N=1,098	0.426 [0.428] N=871
Stunted, HAZ below -2	0.065 [0.050] N=1,658	0.083 [0.054] N=1,506	0.076 [0.060] N=1,325	**0.124** [0.062] N=1,119**	**0.144* [0.077] N=886**
Underweight, WAZ below -2, through 120 months	0.034 [0.034] N=1,668	0.053 [0.037] N=1,516	0.031 [0.043] N=1,334	**0.085* [0.044] N=1,122**	**0.129** [0.050] N=888**
Thinness, BMIZ below -2	0.016 [0.037] N=1,629	-0.002 [0.041] N=1,476	-0.013 [0.044] N=1,300	0.020 [0.048] N=1,098	-0.007 [0.067] N=871
Overweight including obesity, BMIZ above +1	0.062 [0.070] N=1,629	0.037 [0.080] N=1,476	0.060 [0.090] N=1,300	0.117 [0.106] N=1,098	0.131 [0.127] N=871
Obesity, BMIZ above +2	0.079 [0.054] N=1,629	0.053 [0.058] N=1,476	**0.107* [0.064] N=1,300**	**0.145** [0.066] N=1,098**	0.092 [0.078] N=871

Specification 5 Monthly fixed effects female and number of aid types (0-6) No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Height-for-age z score	-0.290	0.049 [0.211] N=1,855	0.116 [0.223] N=1,712	0.208 [0.234] N=1,539	0.112 [0.248] N=1,347	0.230 [0.270] N=1,126
Weight-for-age z score, through 120 months	0.300	-0.048 [0.152] N=1,869	-0.049 [0.163] N=1,726	0.023 [0.174] N=1,552	0.018 [0.188] N=1,354	0.087 [0.210] N=1,132
BMI-for-age z score	0.569	-0.043 [0.173] N=1,822	-0.062 [0.182] N=1,678	0.003 [0.188] N=1,510	0.003 [0.207] N=1,322	0.042 [0.228] N=1,105
Stunted, HAZ below -2	0.178 P=322	-0.030 [0.038] N=1,855	-0.031 [0.040] N=1,712	-0.044 [0.043] N=1,539	-0.037 [0.044] N=1,347	-0.069 [0.049] N=1,126
Underweight, WAZ below -2, through 120 months	0.048 P=91	**0.045* [0.023] N=1,869**	**0.053** [0.023] N=1,726**	**0.044* [0.025] N=1,552**	**0.069*** [0.025] N=1,354**	**0.070** [0.027] N=1,132**
Thinness, BMIZ below -2	0.085 P=157	0.037 [0.027] N=1,822	0.031 [0.029] N=1,678	0.028 [0.030] N=1,510	0.050 [0.031] N=1,322	0.040 [0.035] N=1,105
Overweight including obesity, BMIZ above +1	0.409 P=748	0.043 [0.050] N=1,822	0.027 [0.054] N=1,678	0.039 [0.057] N=1,510	0.065 [0.062] N=1,322	0.060 [0.066] N=1,105
Obesity, BMIZ above +2	0.219 P=396	0.040 [0.037] N=1,822	0.022 [0.038] N=1,678	0.058 [0.040] N=1,510	0.067 [0.042] N=1,322	0.036 [0.045] N=1,105

Specification 5 Monthly fixed effects female and number of aid types (0-6) 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Height-for-age z score	-0.352 [0.306] N=1,566	-0.306 [0.337] N=1,423	-0.231 [0.372] N=1,250	-0.603 [0.410] N=1,058	-0.620 [0.541] N=837
Weight-for-age z score, through 120 months	-0.161 [0.193] N=1,575	-0.182 [0.217] N=1,432	-0.086 [0.241] N=1,258	-0.193 [0.272] N=1,060	-0.150 [0.353] N=838
BMI-for-age z score	0.047 [0.249] N=1,540	0.034 [0.275] N=1,396	0.183 [0.294] N=1,228	0.235 [0.337] N=1,040	0.317 [0.456] N=823
Stunted, HAZ below -2	0.051 [0.053] N=1,566	0.064 [0.056] N=1,423	0.059 [0.063] N=1,250	0.106 [0.066] N=1,058	0.084 [0.084] N=837
Underweight, WAZ below -2, through 120 months	0.042 [0.036] N=1,575	0.059 [0.038] N=1,432	0.042 [0.044] N=1,258	**0.101** [0.044] N=1,060**	**0.124** [0.052] N=838**
Thinness, BMIZ below -2	0.023 [0.038] N=1,540	0.008 [0.042] N=1,396	-0.002 [0.045] N=1,228	0.031 [0.048] N=1,040	0.022 [0.071] N=823
Overweight including obesity, BMIZ above +1	0.056 [0.073] N=1,540	0.028 [0.083] N=1,396	0.056 [0.093] N=1,228	0.108 [0.108] N=1,040	0.107 [0.134] N=823
Obesity, BMIZ above +2	0.067 [0.053] N=1,540	0.042 [0.058] N=1,396	**0.113* [0.063] N=1,228**	**0.148** [0.065] N=1,040**	0.083 [0.079] N=823

Specification 6 Monthly fixed effects female and any aid (0/1) No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Height-for-age z score	-0.290	0.038 [0.210] N=1,855	0.112 [0.221] N=1,712	0.210 [0.232] N=1,539	0.118 [0.245] N=1,347	0.255 [0.265] N=1,126
Weight-for-age z score, through 120 months	0.300	-0.055 [0.151] N=1,869	-0.051 [0.162] N=1,726	0.023 [0.174] N=1,552	0.022 [0.187] N=1,354	0.098 [0.209] N=1,132
BMI-for-age z score	0.569	-0.034 [0.172] N=1,822	-0.057 [0.180] N=1,678	0.006 [0.188] N=1,510	0.009 [0.206] N=1,322	0.050 [0.228] N=1,105
Stunted, HAZ below -2	0.178 P=322	-0.028 [0.038] N=1,855	-0.030 [0.039] N=1,712	-0.045 [0.042] N=1,539	-0.039 [0.043] N=1,347	-0.074 [0.048] N=1,126
Underweight, WAZ below -2, through 120 months	0.048 P=91	**0.043* [0.023] N=1,869**	**0.051** [0.023] N=1,726**	**0.043* [0.025] N=1,552**	**0.066*** [0.025] N=1,354**	**0.067** [0.026] N=1,132**
Thinness, BMIZ below -2	0.085 P=157	0.038 [0.027] N=1,822	0.032 [0.029] N=1,678	0.029 [0.031] N=1,510	0.051 [0.031] N=1,322	0.040 [0.036] N=1,105
Overweight including obesity, BMIZ above +1	0.409 P=748	0.046 [0.050] N=1,822	0.028 [0.054] N=1,678	0.041 [0.057] N=1,510	0.068 [0.062] N=1,322	0.064 [0.066] N=1,105
Obesity, BMIZ above +2	0.219 P=396	0.043 [0.037] N=1,822	0.023 [0.038] N=1,678	0.059 [0.040] N=1,510	0.069 [0.042] N=1,322	0.038 [0.045] N=1,105

Specification 6 Monthly fixed effects female and any aid (0/1) 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Height-for-age z score	-0.368 [0.307] N=1,566	-0.309 [0.335] N=1,423	-0.223 [0.371] N=1,250	-0.580 [0.405] N=1,058	-0.540 [0.525] N=837
Weight-for-age z score, through 120 months	-0.168 [0.190] N=1,575	-0.180 [0.216] N=1,432	-0.079 [0.241] N=1,258	-0.169 [0.272] N=1,060	-0.110 [0.356] N=838
BMI-for-age z score	0.066 [0.248] N=1,540	0.044 [0.274] N=1,396	0.192 [0.292] N=1,228	0.255 [0.338] N=1,040	0.339 [0.460] N=823
Stunted, HAZ below -2	0.054 [0.052] N=1,566	0.064 [0.056] N=1,423	0.056 [0.063] N=1,250	0.098 [0.066] N=1,058	0.065 [0.083] N=837
Underweight, WAZ below -2, through 120 months	0.038 [0.035] N=1,575	0.056 [0.038] N=1,432	0.038 [0.044] N=1,258	**0.093** [0.045] N=1,060**	**0.112** [0.052] N=838**
Thinness, BMIZ below -2	0.024 [0.038] N=1,540	0.010 [0.042] N=1,396	-0.000 [0.045] N=1,228	0.033 [0.048] N=1,040	0.024 [0.070] N=823
Overweight including obesity, BMIZ above +1	0.062 [0.074] N=1,540	0.033 [0.084] N=1,396	0.062 [0.093] N=1,228	0.120 [0.110] N=1,040	0.125 [0.138] N=823
Obesity, BMIZ above +2	0.071 [0.053] N=1,540	0.043 [0.058] N=1,396	**0.114* [0.062] N=1,228**	**0.152** [0.065] N=1,040**	0.087 [0.080] N=823

Mother RDD secondary analysis

Specification 1 Monthly child-age fixed effects No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Height-for-age z score	-0.181	0.029 [0.136] N=2,410	0.104 [0.143] N=2,198	0.113 [0.152] N=1,952	-0.011 [0.162] N=1,680	0.053 [0.188] N=1,357
Weight-for-age z score, through 120 months	0.371	-0.028 [0.101] N=2,402	0.019 [0.105] N=2,199	0.098 [0.115] N=1,951	0.104 [0.118] N=1,676	0.087 [0.124] N=1,356
BMI-for-age z score	0.572	0.193 [0.132] N=2,368	0.218 [0.140] N=2,160	**0.310** [0.153] N=1,916**	**0.359** [0.159] N=1,647**	**0.301* [0.176] N=1,328**
Stunted, HAZ below -2	0.166 P=389	0.010 [0.030] N=2,410	-0.007 [0.032] N=2,198	0.001 [0.035] N=1,952	0.017 [0.038] N=1,680	-0.009 [0.042] N=1,357
Underweight, WAZ below -2, through 120 months	0.043 P=103	0.014 [0.016] N=2,402	0.005 [0.017] N=2,199	-0.003 [0.018] N=1,951	0.001 [0.019] N=1,676	0.013 [0.022] N=1,356
Thinness, BMIZ below -2	0.086 P=206	-0.005 [0.019] N=2,368	-0.005 [0.020] N=2,160	-0.010 [0.022] N=1,916	-0.021 [0.023] N=1,647	-0.020 [0.025] N=1,328
Overweight including obesity, BMIZ above +1	0.415 P=978	0.045 [0.039] N=2,368	0.042 [0.041] N=2,160	0.049 [0.045] N=1,916	0.048 [0.049] N=1,647	0.019 [0.053] N=1,328
Obesity, BMIZ above +2	0.217 P=505	0.010 [0.032] N=2,368	0.021 [0.033] N=2,160	0.050 [0.036] N=1,916	0.060 [0.039] N=1,647	0.024 [0.044] N=1,328

Specification 1 Monthly child-age fixed effects 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Height-for-age z score	0.054 [0.174] N=2,045	0.212 [0.183] N=1,833	0.247 [0.209] N=1,587	0.028 [0.235] N=1,315	0.205 [0.292] N=992
Weight-for-age z score, through 120 months	-0.172 [0.157] N=2,036	-0.111 [0.173] N=1,833	0.005 [0.205] N=1,585	0.002 [0.241] N=1,310	-0.171 [0.292] N=990
BMI-for-age z score	0.068 [0.200] N=2,008	0.079 [0.223] N=1,800	0.224 [0.266] N=1,556	0.311 [0.307] N=1,287	0.036 [0.397] N=968
Stunted, HAZ below -2	0.014 [0.038] N=2,045	-0.016 [0.043] N=1,833	-0.007 [0.049] N=1,587	0.033 [0.062] N=1,315	-0.031 [0.070] N=992
Underweight, WAZ below -2, through 120 months	0.027 [0.023] N=2,036	0.011 [0.026] N=1,833	-0.005 [0.027] N=1,585	0.004 [0.031] N=1,310	0.031 [0.042] N=990
Thinness, BMIZ below -2	0.004 [0.028] N=2,008	0.003 [0.030] N=1,800	-0.004 [0.035] N=1,556	-0.029 [0.039] N=1,287	-0.030 [0.048] N=968
Overweight including obesity, BMIZ above +1	0.059 [0.055] N=2,008	0.052 [0.061] N=1,800	0.068 [0.071] N=1,556	0.071 [0.084] N=1,287	-0.030 [0.102] N=968
Obesity, BMIZ above +2	-0.028 [0.043] N=2,008	-0.015 [0.048] N=1,800	0.031 [0.055] N=1,556	0.051 [0.067] N=1,287	-0.082 [0.081] N=968

Specification 2 Monthly fixed effects and female No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Height-for-age z score	-0.181	0.031 [0.135] N=2,410	0.106 [0.142] N=2,198	0.113 [0.152] N=1,952	-0.011 [0.162] N=1,680	0.053 [0.188] N=1,357
Weight-for-age z score, through 120 months	0.371	-0.019 [0.099] N=2,402	0.026 [0.103] N=2,199	0.101 [0.113] N=1,951	0.102 [0.115] N=1,676	0.079 [0.119] N=1,356
BMI-for-age z score	0.572	0.202 [0.133] N=2,368	0.224 [0.141] N=2,160	**0.312** [0.154] N=1,916**	**0.356** [0.160] N=1,647**	**0.293* [0.176] N=1,328**
Stunted, HAZ below -2	0.166 P=389	0.010 [0.030] N=2,410	-0.006 [0.032] N=2,198	0.001 [0.035] N=1,952	0.017 [0.038] N=1,680	-0.009 [0.043] N=1,357
Underweight, WAZ below -2, through 120 months	0.043 P=103	0.014 [0.016] N=2,402	0.005 [0.017] N=2,199	-0.004 [0.018] N=1,951	0.002 [0.019] N=1,676	0.013 [0.022] N=1,356
Thinness, BMIZ below -2	0.086 P=206	-0.004 [0.019] N=2,368	-0.005 [0.020] N=2,160	-0.010 [0.022] N=1,916	-0.021 [0.023] N=1,647	-0.020 [0.025] N=1,328
Overweight including obesity, BMIZ above +1	0.415 P=978	0.048 [0.039] N=2,368	0.044 [0.041] N=2,160	0.050 [0.045] N=1,916	0.048 [0.049] N=1,647	0.016 [0.053] N=1,328
Obesity, BMIZ above +2	0.217 P=505	0.014 [0.032] N=2,368	0.024 [0.033] N=2,160	0.050 [0.036] N=1,916	0.059 [0.040] N=1,647	0.021 [0.044] N=1,328

Specification 2 Monthly fixed effects and female 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Height-for-age z score	0.052 [0.172] N=2,045	0.211 [0.181] N=1,833	0.239 [0.207] N=1,587	0.021 [0.233] N=1,315	0.190 [0.291] N=992
Weight-for-age z score, through 120 months	-0.148 [0.156] N=2,036	-0.084 [0.171] N=1,833	0.026 [0.204] N=1,585	0.023 [0.238] N=1,310	-0.134 [0.288] N=990
BMI-for-age z score	0.098 [0.203] N=2,008	0.108 [0.226] N=1,800	0.254 [0.270] N=1,556	0.343 [0.311] N=1,287	0.079 [0.397] N=968
Stunted, HAZ below -2	0.017 [0.038] N=2,045	-0.014 [0.043] N=1,833	-0.005 [0.049] N=1,587	0.037 [0.062] N=1,315	-0.025 [0.070] N=992
Underweight, WAZ below -2, through 120 months	0.027 [0.023] N=2,036	0.010 [0.026] N=1,833	-0.005 [0.027] N=1,585	0.003 [0.031] N=1,310	0.029 [0.043] N=990
Thinness, BMIZ below -2	0.007 [0.028] N=2,008	0.005 [0.030] N=1,800	-0.004 [0.035] N=1,556	-0.029 [0.039] N=1,287	-0.033 [0.049] N=968
Overweight including obesity, BMIZ above +1	0.070 [0.056] N=2,008	0.062 [0.062] N=1,800	0.076 [0.072] N=1,556	0.080 [0.085] N=1,287	-0.017 [0.102] N=968
Obesity, BMIZ above +2	-0.015 [0.044] N=2,008	-0.002 [0.049] N=1,800	0.042 [0.056] N=1,556	0.064 [0.068] N=1,287	-0.066 [0.081] N=968

Specification 3 Monthly fixed effects female and household income groups No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Height-for-age z score	-0.181	0.031 [0.130] N=2,410	0.098 [0.139] N=2,198	0.101 [0.148] N=1,952	-0.025 [0.156] N=1,680	0.046 [0.183] N=1,357
Weight-for-age z score, through 120 months	0.371	-0.012 [0.097] N=2,402	0.027 [0.101] N=2,199	0.094 [0.111] N=1,951	0.097 [0.113] N=1,676	0.071 [0.117] N=1,356
BMI-for-age z score	0.572	0.206 [0.133] N=2,368	0.229 [0.141] N=2,160	**0.315** [0.155] N=1,916**	**0.364** [0.160] N=1,647**	**0.296* [0.176] N=1,328**
Stunted, HAZ below -2	0.166 P=389	0.010 [0.029] N=2,410	-0.006 [0.032] N=2,198	0.002 [0.034] N=1,952	0.020 [0.037] N=1,680	-0.009 [0.041] N=1,357
Underweight, WAZ below -2, through 120 months	0.043 P=103	0.013 [0.016] N=2,402	0.005 [0.017] N=2,199	-0.003 [0.018] N=1,951	0.002 [0.019] N=1,676	0.014 [0.022] N=1,356
Thinness, BMIZ below -2	0.086 P=206	-0.005 [0.019] N=2,368	-0.005 [0.020] N=2,160	-0.011 [0.022] N=1,916	-0.021 [0.024] N=1,647	-0.020 [0.025] N=1,328
Overweight including obesity, BMIZ above +1	0.415 P=978	0.049 [0.039] N=2,368	0.045 [0.041] N=2,160	0.049 [0.045] N=1,916	0.048 [0.049] N=1,647	0.014 [0.053] N=1,328
Obesity, BMIZ above +2	0.217 P=505	0.015 [0.032] N=2,368	0.026 [0.033] N=2,160	0.053 [0.036] N=1,916	0.062 [0.039] N=1,647	0.022 [0.044] N=1,328

Specification 3 Monthly fixed effects female and household income groups 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Height-for-age z score	0.050 [0.172] N=2,045	0.199 [0.182] N=1,833	0.209 [0.207] N=1,587	-0.011 [0.232] N=1,315	0.150 [0.294] N=992
Weight-for-age z score, through 120 months	-0.140 [0.153] N=2,036	-0.085 [0.169] N=1,833	0.008 [0.202] N=1,585	0.006 [0.231] N=1,310	-0.169 [0.281] N=990
BMI-for-age z score	0.099 [0.204] N=2,008	0.110 [0.227] N=1,800	0.257 [0.273] N=1,556	0.350 [0.309] N=1,287	0.082 [0.394] N=968
Stunted, HAZ below -2	0.016 [0.038] N=2,045	-0.013 [0.042] N=1,833	-0.001 [0.048] N=1,587	0.043 [0.061] N=1,315	-0.021 [0.070] N=992
Underweight, WAZ below -2, through 120 months	0.026 [0.022] N=2,036	0.010 [0.026] N=1,833	-0.004 [0.028] N=1,585	0.004 [0.032] N=1,310	0.031 [0.045] N=990
Thinness, BMIZ below -2	0.006 [0.028] N=2,008	0.004 [0.030] N=1,800	-0.006 [0.035] N=1,556	-0.031 [0.039] N=1,287	-0.038 [0.049] N=968
Overweight including obesity, BMIZ above +1	0.070 [0.056] N=2,008	0.063 [0.062] N=1,800	0.073 [0.071] N=1,556	0.077 [0.084] N=1,287	-0.024 [0.101] N=968
Obesity, BMIZ above +2	-0.014 [0.044] N=2,008	-0.003 [0.049] N=1,800	0.041 [0.055] N=1,556	0.065 [0.066] N=1,287	-0.069 [0.082] N=968

Specification 4 Monthly fixed effects female and per-person income groups No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Height-for-age z score	-0.181	0.012 [0.128] N=2,410	0.076 [0.136] N=2,198	0.077 [0.144] N=1,952	-0.042 [0.155] N=1,680	0.001 [0.183] N=1,357
Weight-for-age z score, through 120 months	0.371	-0.021 [0.094] N=2,402	0.019 [0.098] N=2,199	0.088 [0.108] N=1,951	0.091 [0.110] N=1,676	0.053 [0.112] N=1,356
BMI-for-age z score	0.572	0.204 [0.133] N=2,368	0.229 [0.141] N=2,160	**0.318** [0.155] N=1,916**	**0.363** [0.160] N=1,647**	**0.298* [0.177] N=1,328**
Stunted, HAZ below -2	0.166 P=389	0.014 [0.029] N=2,410	-0.000 [0.031] N=2,198	0.008 [0.034] N=1,952	0.022 [0.037] N=1,680	-0.000 [0.041] N=1,357
Underweight, WAZ below -2, through 120 months	0.043 P=103	0.014 [0.016] N=2,402	0.006 [0.017] N=2,199	-0.003 [0.018] N=1,951	0.002 [0.019] N=1,676	0.014 [0.021] N=1,356
Thinness, BMIZ below -2	0.086 P=206	-0.004 [0.019] N=2,368	-0.005 [0.020] N=2,160	-0.011 [0.022] N=1,916	-0.022 [0.023] N=1,647	-0.019 [0.024] N=1,328
Overweight including obesity, BMIZ above +1	0.415 P=978	0.048 [0.039] N=2,368	0.045 [0.042] N=2,160	0.051 [0.046] N=1,916	0.049 [0.049] N=1,647	0.018 [0.054] N=1,328
Obesity, BMIZ above +2	0.217 P=505	0.014 [0.031] N=2,368	0.025 [0.033] N=2,160	0.051 [0.036] N=1,916	0.059 [0.039] N=1,647	0.021 [0.044] N=1,328

Specification 4 Monthly fixed effects female and per-person income groups 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Height-for-age z score	0.057 [0.172] N=2,045	0.207 [0.183] N=1,833	0.231 [0.208] N=1,587	0.032 [0.238] N=1,315	0.157 [0.303] N=992
Weight-for-age z score, through 120 months	-0.138 [0.151] N=2,036	-0.081 [0.166] N=1,833	0.023 [0.199] N=1,585	0.032 [0.227] N=1,310	-0.153 [0.268] N=990
BMI-for-age z score	0.096 [0.205] N=2,008	0.105 [0.228] N=1,800	0.251 [0.272] N=1,556	0.338 [0.311] N=1,287	0.073 [0.394] N=968
Stunted, HAZ below -2	0.015 [0.038] N=2,045	-0.013 [0.042] N=1,833	-0.004 [0.048] N=1,587	0.031 [0.061] N=1,315	-0.023 [0.070] N=992
Underweight, WAZ below -2, through 120 months	0.026 [0.022] N=2,036	0.010 [0.026] N=1,833	-0.006 [0.027] N=1,585	0.002 [0.031] N=1,310	0.028 [0.043] N=990
Thinness, BMIZ below -2	0.009 [0.028] N=2,008	0.006 [0.029] N=1,800	-0.002 [0.035] N=1,556	-0.028 [0.038] N=1,287	-0.031 [0.046] N=968
Overweight including obesity, BMIZ above +1	0.070 [0.056] N=2,008	0.062 [0.062] N=1,800	0.077 [0.071] N=1,556	0.079 [0.085] N=1,287	-0.017 [0.103] N=968
Obesity, BMIZ above +2	-0.015 [0.044] N=2,008	-0.003 [0.049] N=1,800	0.041 [0.055] N=1,556	0.062 [0.067] N=1,287	-0.069 [0.080] N=968

Specification 5 Monthly fixed effects female and number of aid types (0-6) No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Height-for-age z score	-0.195	0.092 [0.133] N=2,272	0.163 [0.140] N=2,071	0.170 [0.150] N=1,836	0.039 [0.160] N=1,577	0.087 [0.185] N=1,279
Weight-for-age z score, through 120 months	0.378	0.002 [0.103] N=2,265	0.040 [0.108] N=2,073	0.129 [0.119] N=1,835	0.129 [0.122] N=1,574	0.125 [0.127] N=1,278
BMI-for-age z score	0.590	0.198 [0.134] N=2,233	0.233 [0.143] N=2,035	**0.324** [0.157] N=1,802**	**0.343** [0.162] N=1,547**	0.294 [0.182] N=1,253
Stunted, HAZ below -2	0.168 P=370	0.009 [0.030] N=2,272	-0.007 [0.032] N=2,071	0.004 [0.035] N=1,836	0.023 [0.038] N=1,577	-0.000 [0.042] N=1,279
Underweight, WAZ below -2, through 120 months	0.043 P=98	0.012 [0.016] N=2,265	0.004 [0.018] N=2,073	-0.006 [0.018] N=1,835	-0.001 [0.019] N=1,574	0.009 [0.022] N=1,278
Thinness, BMIZ below -2	0.084 P=191	-0.005 [0.019] N=2,233	-0.009 [0.019] N=2,035	-0.013 [0.021] N=1,802	-0.018 [0.022] N=1,547	-0.019 [0.024] N=1,253
Overweight including obesity, BMIZ above +1	0.419 P=930	0.046 [0.039] N=2,233	0.045 [0.042] N=2,035	0.053 [0.045] N=1,802	0.052 [0.048] N=1,547	0.023 [0.053] N=1,253
Obesity, BMIZ above +2	0.219 P=482	0.006 [0.032] N=2,233	0.017 [0.034] N=2,035	0.042 [0.036] N=1,802	0.048 [0.040] N=1,547	0.012 [0.045] N=1,253

Specification 5 Monthly fixed effects female and number of aid types (0-6) 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Height-for-age z score	0.124 [0.182] N=1,924	0.281 [0.193] N=1,723	0.299 [0.220] N=1,488	0.079 [0.255] N=1,229	0.174 [0.311] N=931
Weight-for-age z score, through 120 months	-0.118 [0.163] N=1,915	-0.067 [0.180] N=1,723	0.076 [0.215] N=1,485	0.081 [0.247] N=1,224	-0.007 [0.293] N=928
BMI-for-age z score	0.108 [0.211] N=1,889	0.140 [0.238] N=1,691	0.315 [0.281] N=1,458	0.366 [0.319] N=1,203	0.173 [0.408] N=909
Stunted, HAZ below -2	0.006 [0.040] N=1,924	-0.027 [0.045] N=1,723	-0.012 [0.050] N=1,488	0.032 [0.062] N=1,229	-0.022 [0.070] N=931
Underweight, WAZ below -2, through 120 months	0.023 [0.024] N=1,915	0.006 [0.027] N=1,723	-0.010 [0.028] N=1,485	-0.001 [0.032] N=1,224	0.018 [0.045] N=928
Thinness, BMIZ below -2	0.002 [0.029] N=1,889	-0.007 [0.032] N=1,691	-0.015 [0.039] N=1,458	-0.029 [0.042] N=1,203	-0.039 [0.055] N=909
Overweight including obesity, BMIZ above +1	0.061 [0.058] N=1,889	0.055 [0.065] N=1,691	0.078 [0.074] N=1,458	0.086 [0.087] N=1,203	0.007 [0.106] N=909
Obesity, BMIZ above +2	-0.028 [0.044] N=1,889	-0.015 [0.050] N=1,691	0.029 [0.056] N=1,458	0.041 [0.069] N=1,203	-0.083 [0.081] N=909

Specification 6 Monthly fixed effects female and any aid (0/1) No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Height-for-age z score	-0.195	0.090 [0.135] N=2,272	0.163 [0.141] N=2,071	0.167 [0.151] N=1,836	0.038 [0.160] N=1,577	0.088 [0.186] N=1,279
Weight-for-age z score, through 120 months	0.378	0.003 [0.103] N=2,265	0.043 [0.108] N=2,073	0.133 [0.119] N=1,835	0.130 [0.122] N=1,574	0.126 [0.126] N=1,278
BMI-for-age z score	0.590	0.206 [0.135] N=2,233	**0.241* [0.144] N=2,035**	**0.339** [0.158] N=1,802**	**0.356** [0.163] N=1,547**	**0.318* [0.181] N=1,253**
Stunted, HAZ below -2	0.168 P=370	0.009 [0.030] N=2,272	-0.007 [0.032] N=2,071	0.004 [0.035] N=1,836	0.024 [0.037] N=1,577	-0.002 [0.041] N=1,279
Underweight, WAZ below -2, through 120 months	0.043 P=98	0.012 [0.016] N=2,265	0.003 [0.018] N=2,073	-0.005 [0.018] N=1,835	0.001 [0.020] N=1,574	0.010 [0.022] N=1,278
Thinness, BMIZ below -2	0.084 P=191	-0.005 [0.019] N=2,233	-0.009 [0.019] N=2,035	-0.014 [0.021] N=1,802	-0.018 [0.021] N=1,547	-0.020 [0.023] N=1,253
Overweight including obesity, BMIZ above +1	0.419 P=930	0.048 [0.040] N=2,233	0.047 [0.042] N=2,035	0.058 [0.046] N=1,802	0.057 [0.049] N=1,547	0.030 [0.054] N=1,253
Obesity, BMIZ above +2	0.219 P=482	0.008 [0.032] N=2,233	0.019 [0.034] N=2,035	0.046 [0.036] N=1,802	0.052 [0.040] N=1,547	0.017 [0.044] N=1,253

Specification 6 Monthly fixed effects female and any aid (0/1) 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Height-for-age z score	0.114 [0.182] N=1,924	0.270 [0.192] N=1,723	0.286 [0.219] N=1,488	0.045 [0.251] N=1,229	0.159 [0.304] N=931
Weight-for-age z score, through 120 months	-0.129 [0.163] N=1,915	-0.077 [0.179] N=1,723	0.068 [0.216] N=1,485	0.057 [0.246] N=1,224	-0.029 [0.290] N=928
BMI-for-age z score	0.102 [0.210] N=1,889	0.135 [0.237] N=1,691	0.314 [0.280] N=1,458	0.358 [0.316] N=1,203	0.172 [0.401] N=909
Stunted, HAZ below -2	0.009 [0.039] N=1,924	-0.023 [0.044] N=1,723	-0.007 [0.050] N=1,488	0.043 [0.062] N=1,229	-0.013 [0.070] N=931
Underweight, WAZ below -2, through 120 months	0.024 [0.024] N=1,915	0.007 [0.027] N=1,723	-0.009 [0.028] N=1,485	0.001 [0.032] N=1,224	0.018 [0.045] N=928
Thinness, BMIZ below -2	0.003 [0.029] N=1,889	-0.006 [0.032] N=1,691	-0.015 [0.038] N=1,458	-0.029 [0.042] N=1,203	-0.044 [0.053] N=909
Overweight including obesity, BMIZ above +1	0.060 [0.058] N=1,889	0.054 [0.064] N=1,691	0.078 [0.074] N=1,458	0.086 [0.086] N=1,203	0.002 [0.105] N=909
Obesity, BMIZ above +2	-0.028 [0.044] N=1,889	-0.015 [0.050] N=1,691	0.029 [0.056] N=1,458	0.042 [0.069] N=1,203	-0.084 [0.080] N=909

Notes: Cells report the reform coefficient, clustered standard error in brackets, and N; bold stars indicate *** p<0.01, ** p<0.05, * p<0.10. Five columns use 8- to 4-year parental birth-month bandwidths. The 12-month donut omits months -12 through +11. All specifications include exact child-month fixed effects, split linear trends, and FAKTOR_FERT survey weights. Specification 2 adds female; specifications 3 and 4 separately add household and per-person income groups; specifications 5 and 6 separately add aid count and any aid. WHO 2007 HAZ and BMIZ are valid at months 61-228; WAZ ends at month 120. Implausible z scores are excluded. P is the positive-case count in the eight-year no-donut sample. NA indicates zero standard error. School-level and measurement selection affect interpretation.